The first law

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Learning target: Use a consistent sign convention in an energy balance.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 4.*
Q=mL=1500 J.

02. Worked example: explain the model
90 J in, 30 J out as work, 60 J increase; signs follow stated convention.

03. Read and interpret the model
ΔU=100-40=60 J.

04. Apply the physics
60 J increase.

05. Explain your reasoning
Must know whether the system does work or has work done on it; the energy effect differs.

06. Review: connect an earlier idea *Review: Unit 4, Day 17.*
Inelastic; 4 J can become deformation, thermal energy or sound. Total energy is conserved.

07. Exit check
+20 J.

08. STRETCH: test the model
Examples 60 in and 20 out; 100 in and 60 out. Difference 40 J.

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01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 4.*
$L=|Q|/m\approx 3.333\times 10^5$ J/kg; sample Q=-100000 J.

02. Worked example: explain the model
90 J in, 30 J out as work, 60 J increase; signs follow stated convention.

03. Read and interpret the model
$\Delta U=100-40=60$ J.

04. Apply the physics
$Q=-50$ J; $W_{by}=-60$ J; $\Delta U=+10$ J.

05. Explain your reasoning
Must know whether the system does work or has work done on it; the energy effect differs.

06. Review: connect an earlier idea *Review: Unit 4, Day 17.*
Momentum 48 kg·m/s before and after. $K_i=288$ J; $K_f=148$ J. Inelastic; 140 J leaves kinetic energy.

07. Exit check
Q=10 J; enters.

08. STRETCH: test the model
Q=-20, W=-50,ΔU=+30 J. Reversal gives Q=+20, W=+50,ΔU=-30 J.